

The Phragmén-Lindelöf Theorem and Modular Elliptic Curves

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1. Introduction

In 1957, Rademacher [4,5] wrote two papers on the Phragmén-Lindelöf theorem. The significance of these papers is that they provide (to date) precise inequalities satisfied by a general L -function inside the critical strip. For the Riemann zeta function or for the classical Dirichlet L -functions, special methods of analytic number theory give better estimates than the convexity argument supplied by the Phragmén-Lindelöf theorem. However, in the absence of such special methods, as in the case of certain Artin L -functions or L -functions of elliptic curves, Rademacher's work gives sufficiently accurate estimates useful in many applications (see for example [3]).

Ever since G. Frey [1] had the brilliant idea of linking Fermat's "Last Theorem" with the Taniyama conjecture (that every elliptic curve over $\mathbb{Q}$ is modular), a variety of conjectures have arisen in the theory of elliptic curves. Two of these conjectures are the height and degree conjectures (see section 3 for definitions). The purpose of this paper is to show that for modular elliptic curves, these two conjectures are equivalent. Since Frey had shown the equivalence of the height conjecture and the famous ABC conjecture, our result establishes the equivalence of the degree and ABC conjectures, modulo the Taniyama conjecture. Our main tool will be Rademacher's version of the Phragmén-Lindelöf theorem.

After stating the form of Rademacher's theorem in a slightly general context, we review the relevant facts and conjectures in the theory of elliptic curves. Our main theorems appear in section 4.

1991 *Mathematics Subject Classification.* 11G05, 14H52, 30C80.

¹ Research partially supported by an NSERC Postdoctoral Fellowship.

² Research partially supported by NSERC and FCAR grants.

This paper is in final form and no version of it will be submitted for publication elsewhere.

Acknowledgement. It is a pleasure to thank G. Frey without whose encouragement this paper would not have been written.

2. Rademacher's Theorem

Let

$$L(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

be a Dirichlet series absolutely convergent for $\Re(s) > 1$ and admitting

(i) an Euler product : for $\Re(s) > 1$

$$L(s) = \prod_p L_p(s)$$

where $L_p(s)$ is the inverse of a polynomial of degree d_p in p^{-s} ;

(ii) a positive integer d such that $d_p \leq d$ for all p and equality holds for all but finitely many p ;

(iii) an analytic continuation to an entire function and satisfying a functional equation of the form :

$$\Lambda(s) = w\Lambda^*(1-s)$$

where $w \in \mathbb{C}, |w| = 1$ and

$$\Lambda(s) = A^{s/2} \prod_{i=1}^m \Gamma(\alpha_i s + r_i) L(s)$$

and

$$\Lambda^*(s) = A^{s/2} \prod_{i=1}^m \Gamma(\alpha_i s + r_i) L^*(s)$$

with

$$L_p(s) = \prod_{i=1}^{d_p} (1 - \alpha_i p^{-s}), \quad |\alpha_i| = 1$$

$$L_p^*(s) = \prod_{i=1}^{d_p} (1 - \overline{\alpha_i} p^{-s})$$

and we have set

$$L^*(s) = \prod_p L_p^*(s).$$

Here $A > 0$, $\alpha_i, r_i \in \mathbb{R}$ and m is a natural integer.

Then Rademacher's version of the Phragmén-Lindelöf theorem immediately gives :

PROPOSITION. For $0 \leq \sigma \leq 1$,

$$|L(\sigma + it)| \ll \left(A(|t| + 2)^d\right)^{(1-\sigma)/2} \left(\log(A(|t| + 2)^d)\right)^d$$

for any $L(s)$ satisfying (i), (ii) and (iii) above. The implied constants depend only on the r_i 's.

If $f(z)$ is a normalized new form of weight 2 on $\Gamma_0(N)$, let $f(z) = \sum_{n=1}^{\infty} a_n e^{2\pi i n z}$ be its Fourier expansion at $i\infty$. Writing $a_p/\sqrt{p} = \alpha_p + \overline{\alpha_p}$ with $|\alpha_p| = 1$, the symmetric square L -function associated to f is defined by

$$L_p(s, \text{Sym}^2(f)) = \left(1 - \frac{\alpha_p^2}{p^s}\right)^{-1} \left(1 - \frac{\alpha_p \overline{\alpha_p}}{p^s}\right)^{-1} \left(1 - \frac{\overline{\alpha_p}^2}{p^s}\right)^{-1}$$

for $p \nmid N$ and for $p|N$, it can be defined appropriately (see [9]) and we set

$$L(s, \text{Sym}^2(f)) = \prod_p L_p(s),$$

which converges for $\Re(s) > 1$. By Shimura [9], we know that $L(s, \text{Sym}^2(f))$ extends to an entire function and satisfies a functional equation of the form :

$$\begin{aligned} \Lambda(s) &= A^s \Gamma\left(\frac{s+k-1}{2}\right) \Gamma\left(\frac{s+k}{2}\right) \Gamma\left(\frac{s+1}{2}\right) L(s, \text{Sym}^2(f)) \\ &= w \Lambda(1-s) \end{aligned}$$

where $|w| = 1$, and $\log A = O(\log N)$.

Hence, by Rademacher's version of the Phragmén-Lindelöf, $L(1, \text{Sym}^2(f)) = O((\log N)^3)$.

On the other hand, the Rankin-Selberg method shows that

$$L(1, \text{Sym}^2(f)) = 288\pi^3 \prod_{p^2|N} \left(1 - \frac{1}{p^2}\right) \frac{\langle f, f \rangle}{N}$$

where $\langle, \rangle$ denotes the Petersson inner product. This yields

PROPOSITION. $\log \langle f, f \rangle = O(\log N)$

3. The height and degree conjectures

Let E be an elliptic curve defined over $\mathbb{Q}$ given by a Weierstrass equation :

$$(*) \quad y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6$$

with $a_i \in \mathbb{Z}$. Define

$$\begin{aligned} c_4 &= (a_1^2 + 4a_2)^2 - 24(a_1 a_3 + 2a_4) \\ c_6 &= -(a_1^2 + 4a_2)^2 + 36(a_1^2 + 4a_2)(a_1 a_3 + 2a_4) - 216(a_3^2 + 4a_6) \end{aligned}$$

and

$$\Delta = \frac{c_4^3 - c_6^2}{1728}.$$

(*) is called a minimal model of E if Δ has minimal absolute value compared with all discriminants of Weierstrass equations describing E . A minimal model is unique up to a change of coordinates of the form :

$$\begin{aligned} x &= u^2 x' + r \\ y &= u^3 y' + u^2 s x' + t \end{aligned}$$

with $u, r, s, t \in \mathbb{Z}, u \neq 0$. Consequently, the discriminant of any minimal model is the same. We will denote by Δ_E the discriminant of any minimal model. One knows that Δ_E is an integer. Let N_E be the conductor of E . (See Silverman [10] for a precise definition). One knows that N_E divides Δ_E and has the same prime divisors as Δ_E . For $p \neq 2, 3$, we have $v_p(N_E) \leq 2$ (where $v_p(n)$ is defined such that $p^{-v_p(n)}n$ is coprime to p).

Let N be a positive integer and let $X_0(N)$ be a modular curve of level N . An elliptic curve E over $\mathbb{Q}$ is called modular if there is a non-constant $\mathbb{Q}$ -rational morphism

$$\phi : X_0(N) \longrightarrow E.$$

A famous conjecture of Taniyama (see [10]) predicts that every elliptic curve over $\mathbb{Q}$ is modular. Ribet [7] has show that Taniyama's conjecture implies Fermat's "Last Theorem".

Henceforth, E shall be a modular elliptic curve. Let

$$\omega_E = \frac{dx}{2y + a_1x + a_3}$$

be the Néron differential of E and

$$\omega_E^* = \phi^*(\omega_E)$$

be its pull-back to $X_0(N_E)$.

Then ω_E^* is a holomorphic differential of $X_0(N_E)$ and so by the theory of modular forms,

$$\omega_E^* = \tilde{f}_E(z)dz$$

where $\tilde{f}_E(z)$ is a cusp form of weight 2 on $\Gamma_0(N_E)$. Moreover, by a theorem of Shimura [9] one knows that $\tilde{f}_E(z)$ is an eigenform for all the Hecke operators. Thus $\tilde{f}_E(z) = c_E f_E(z)$ where $f_E(z)$ is a normalized new form of weight 2 and level N_E and c_E is a constant. One knows that $c_E \in \mathbb{Z}$.

There is a result of Raynaud [6] that if N_E is squarefree, $\log |c_E|$ is bounded. Manin conjectures that in fact, $|c_E| = 1$.

The (Faltings) height of E is by definition

$$h(E) = -\frac{1}{2} \log \left(\frac{1}{2\pi} \int_{E(\mathbb{C})} |\omega_E \wedge \overline{\omega_E}| \right).$$

The height conjecture for elliptic curves states that $h(E) = O(\log N_E)$. This conjecture is equivalent to the famous *ABC* conjecture (see Frey [1]).

If E is a modular elliptic curve, the preceding remarks imply that

$$h(E) = -\frac{1}{2} \log \left(\frac{c_E^2}{2\pi \deg \phi} \int_{H^*/\Gamma_0(N)} |f_E|^2 dz \right)$$

where $\deg \phi$ is the degree of the modular parametrization ϕ of E . The degree conjecture is that

$$\log \left(\frac{\deg \phi}{c_E^2} \right) = O \left(\log g(X_0(N_E)) \right),$$

where $g(X_0(N_E))$ is the genus of the modular curve $X_0(N_E)$. We have, for $36 \nmid N_E$ (see [8]),

$$g(X_0(N_E)) = 1 + \frac{\mu}{12} - \frac{\mu_2}{4} - \frac{\mu_3}{3} - \frac{\mu_\infty}{2}$$

where

$$\begin{aligned} \mu &= N_E \prod_{p|N_E} \left(1 + \frac{1}{p}\right) \\ \mu_2 &= \prod_{p|N_E} \left(1 + \left(\frac{-1}{p}\right)\right) \\ \mu_3 &= \prod_{p|N_E} \left(1 + \left(\frac{-3}{p}\right)\right) \\ \mu_\infty &= \sum_{t|N_E} \varphi\left(t, \frac{N_E}{t}\right) \end{aligned}$$

As $\#\{\text{distinct primes dividing } N_E\} \ll \log N_E / \log \log N_E$, we have

$$\log N_E \ll \log g(X_0(N_E)) \ll \log N_E.$$

In the next section, we will show the equivalence of the height and degree conjectures for modular elliptic curves.

4. The main result

From the preceding section, we know

$$h(E) = \frac{1}{2} \log(\deg \phi) - \log |c_E| - \log \langle f_E, f_E \rangle.$$

From the application of the Phragmén-Lindelöf theorem in section 2, we find

$$\log \langle f_E, f_E \rangle = O(\log N_E)$$

so the height and degree conjectures are equivalent. This proves :

THEOREM. *Suppose E is a modular elliptic curve. Then the height and degree conjectures are equivalent.*

COROLLARY. *Under Taniyama's conjecture, the degree conjecture is equivalent to the ABC conjecture.*

REMARK. *Recently, Wiles has announced the semi-stable case of the Taniyama conjecture. This means that when ABC is divisible by 16, the degree conjecture and the ABC conjecture are equivalent.*

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